

Theorem 1 (Intermediate Value Theorem). *If f is a continuous function on the closed interval $[a, b]$, and if d is between $f(a)$ and $f(b)$, then there is a number $c \in [a, b]$ with $f(c) = d$.*

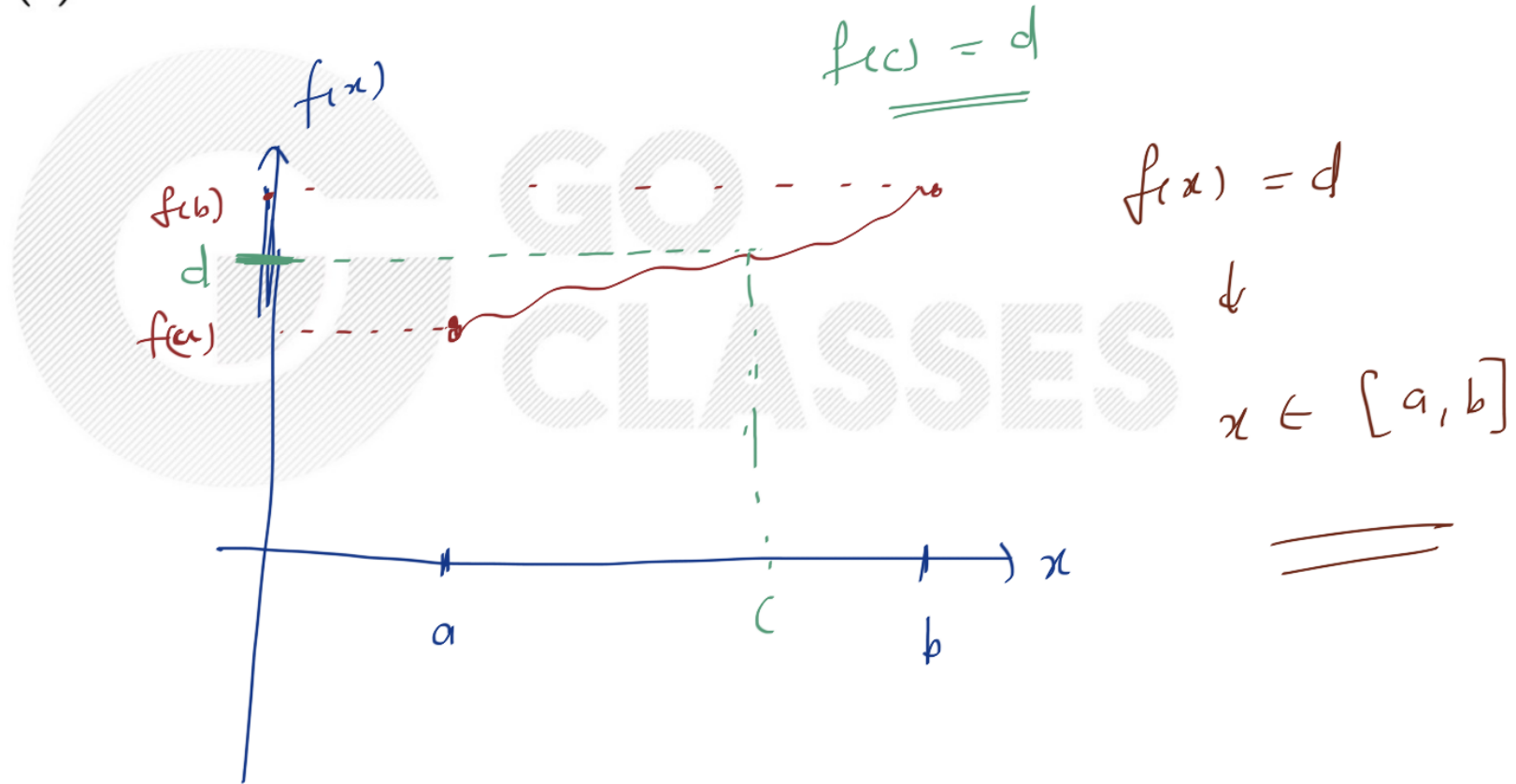
Theorem 2 (Rolle's Theorem). *Suppose f is continuous on $[a, b]$ and differentiable on (a, b) , and suppose that $f(a) = f(b)$. Then there is a number $c \in [a, b]$ with $f'(c) = 0$.*

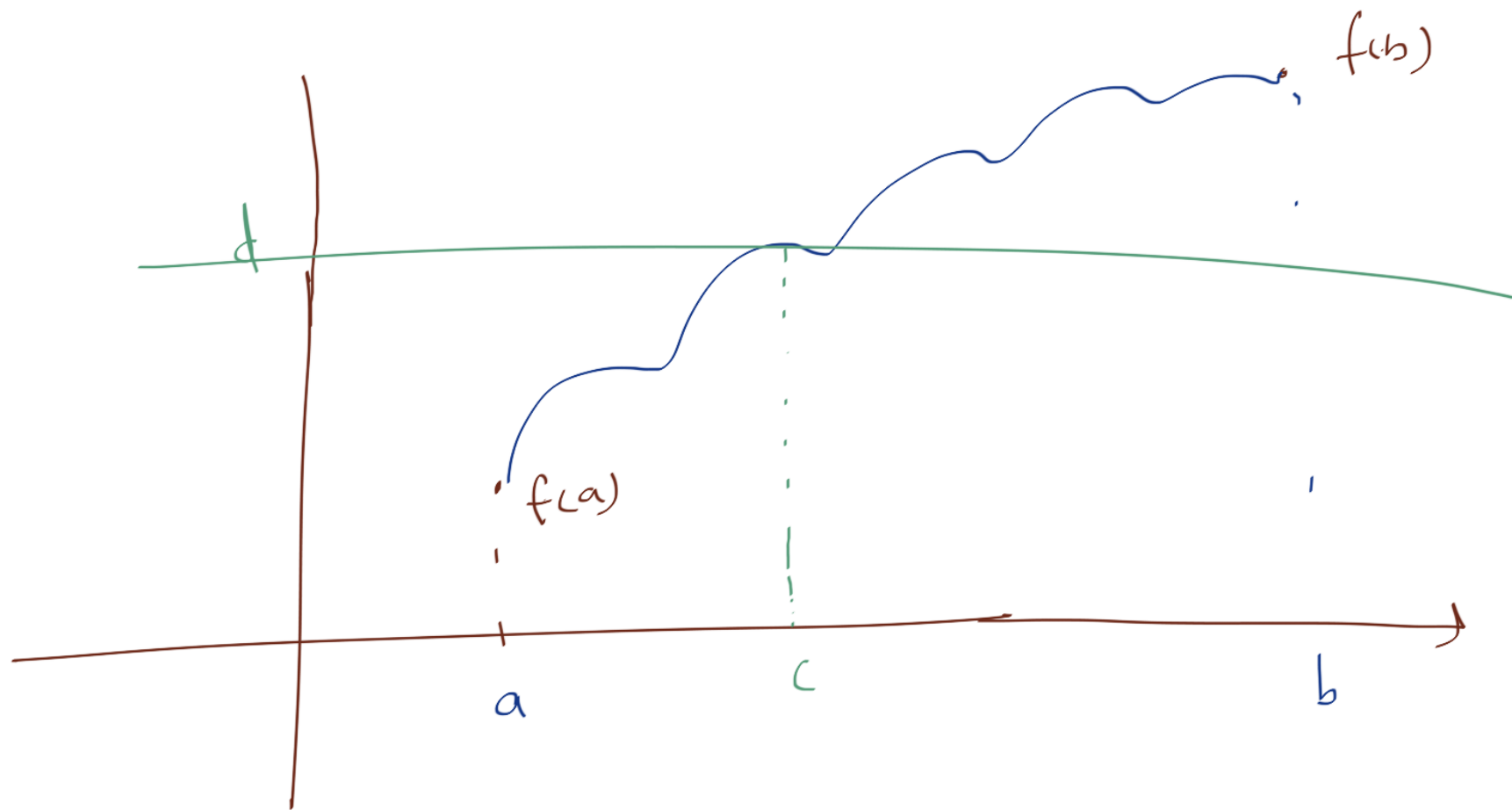
Theorem 3 (The Mean Value Theorem). *Suppose f is continuous on $[a, b]$ and differentiable on (a, b) . Then there is a number $c \in [a, b]$ with*

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$



Theorem 1 (Intermediate Value Theorem). If f is a continuous function on the closed interval $[a, b]$, and if d is between $f(a)$ and $f(b)$, then there is a number $c \in [a, b]$ with $f(c) = d$.





$$\underline{\underline{f(c) = d}}$$

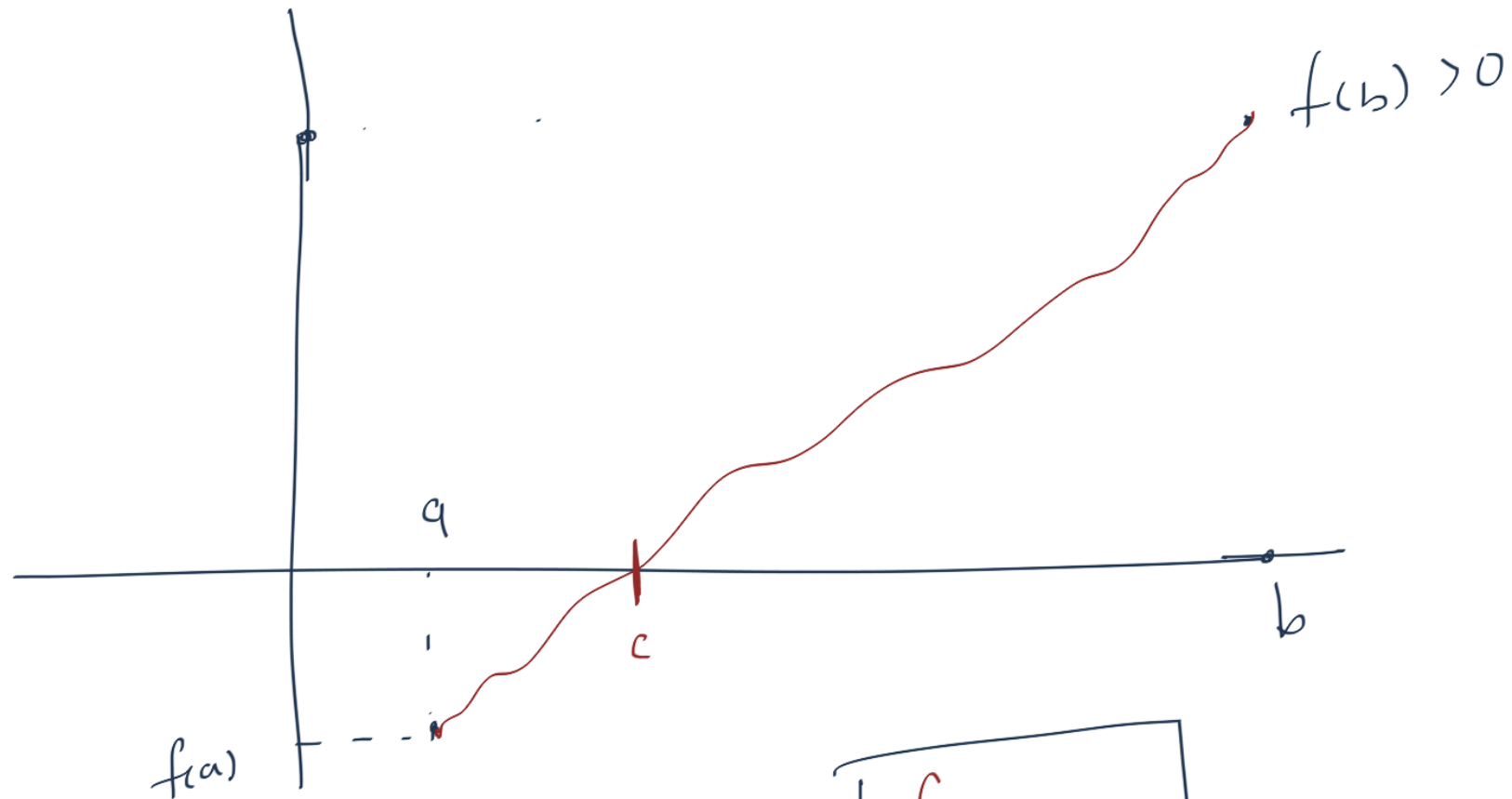
Suppose that $f(x)$ is continuous on the closed interval $[a, b]$
and that N is a number between $f(a)$ and $f(b)$

Then

there must be a number c in the interval $[a, b]$
for which $f(c) = N$.

$\downarrow$ $\uparrow$
this is the same as saying
 $\downarrow$

the equation $f(x) = N$ must have a solution
 $x = c$ in the interval $[a, b]$



$$\boxed{f(c) = 0}$$

Example . Use the Intermediate Value Theorem to show that there is root of the equation

$$4x^3 - 6x^2 + 3x - 2 = 0$$

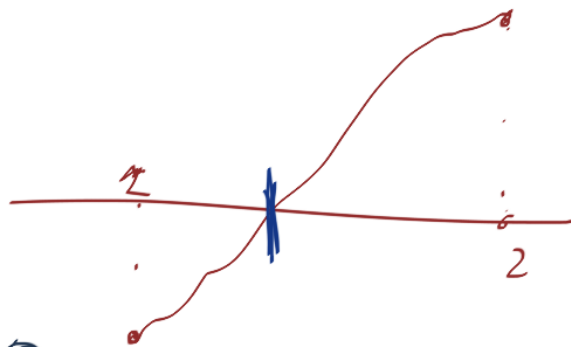
in the interval $[1, 2]$.

$$\begin{cases} f(1) < 0 \\ f(2) > 0 \end{cases}$$

$$f(1) = 4 - 6 + 3 - 2 = -2$$

$$\begin{aligned} f(2) &= 4 \times 8 - 6 \times 4 + 3 \times 2 - 2 \\ &= 32 - 24 + 6 - 2 > 0 \end{aligned}$$

$$38 - 26 = 12 > 0$$



Solution:

Consider the function $f(x) = 4x^3 - 6x^2 + 3x - 2$ over the closed interval $[1, 2]$.

The function f is a polynomial, therefore it is continuous over $[1, 2]$

We have

$$\begin{cases} f(1) &= 4 - 6 + 3 - 2 = -1 \\ f(2) &= 32 - 24 + 6 - 2 = 12 \end{cases}$$

Since:

$$\underline{f(1) < 0 < f(2)}$$

$$\underline{\underline{f(c) = 0}}$$

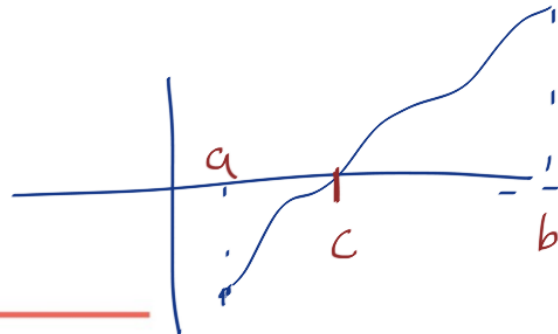
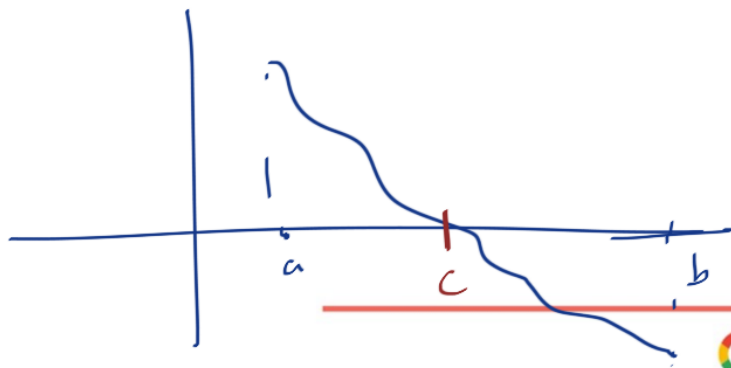
by the Mean-Value-Theorem there exists a value c in the interval $(1, 2)$ such that $f(c) = 0$,
i.e. there is a solution for the equation $f(x) = 0$ in the interval $(1, 2)$.

Example .

$f(x)$ is continuous on $[a, b]$ such that $f(a)f(b) < 0$.

Show the existence of c element of (a, b) such that $f(c) = 0$

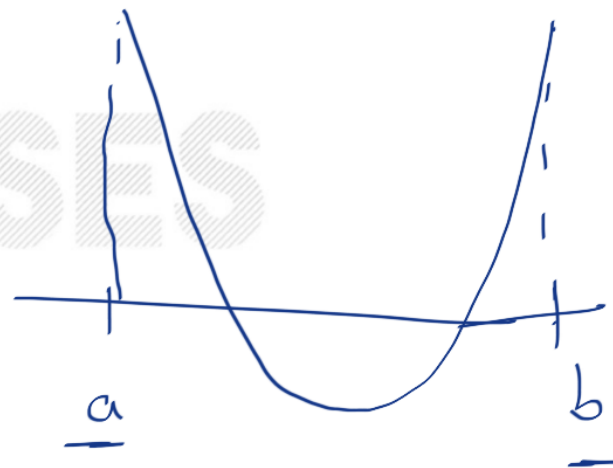
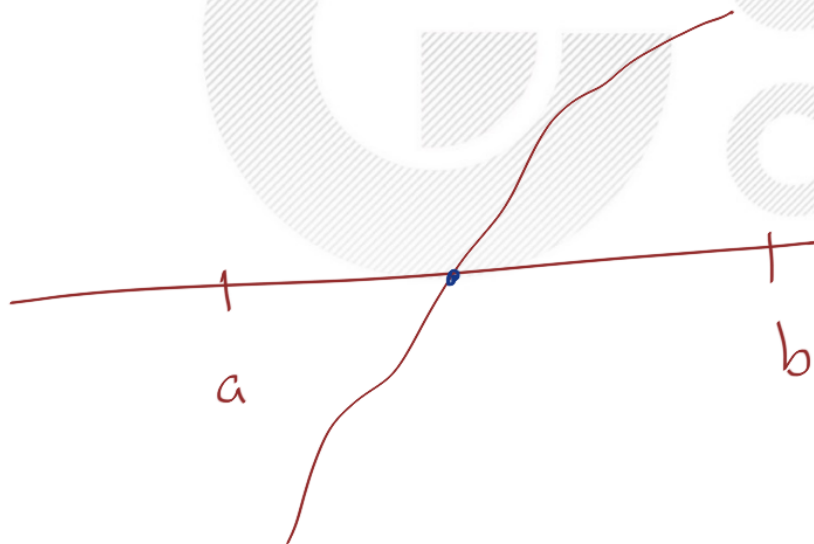
$f(a) \cdot f(b) < 0 \Rightarrow f(a) \text{ \& } f(b) \text{ has opposite sign}$



Example .

Show that the following statement is false by drawing a graph that provides a counterexample:

If $f(x)$ is continuous and has a root in $[a, b]$, then $f(a)$ and $f(b)$ have opposite signs.

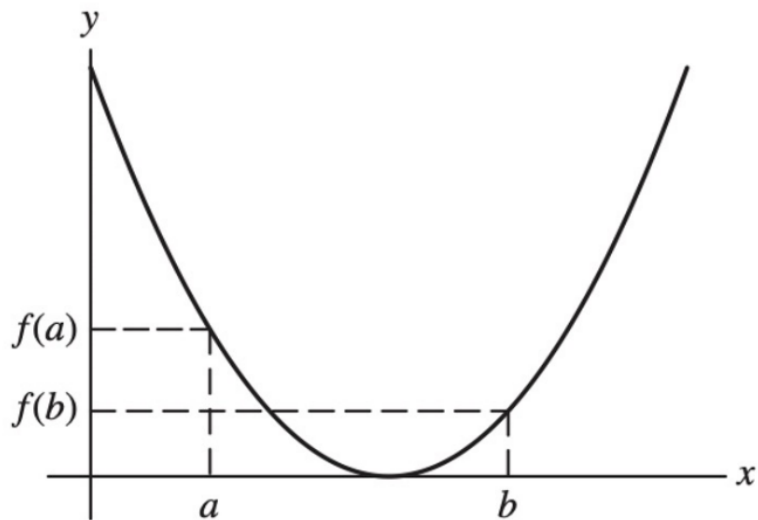


Example .

Show that the following statement is false by drawing a graph that provides a counterexample:

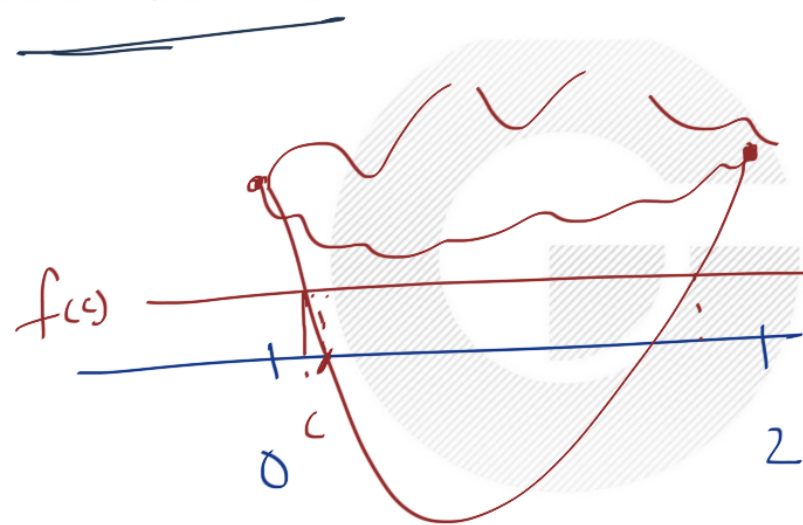
If $f(x)$ is continuous and has a root in $[a, b]$, then $f(a)$ and $f(b)$ have opposite signs.

SOLUTION



Example .

Let $f : [0, 2] \rightarrow \mathbb{R}$ be a continuous function with $f(0) = f(2)$. Show that there is some $c \in [0, 1]$ with $f(c) = f(c+1)$



$$g(x) = f(x) - f(x+1)$$

$$\begin{cases} g(0) = f(0) - f(1) \\ g(1) = f(1) - f(2) \end{cases}$$

$\swarrow$ $f(0)$

$g(x)$ has a root

$$\begin{aligned} f(x) - f(x+1) &= 0 \\ f(x) &= f(x+1) \end{aligned}$$

$g(x)$ is a contⁿ fⁿ and

$$g(0) \cdot g(1) < 0 \quad \Rightarrow \quad g(x) \text{ has a}$$

root b/w $(0,1)$

$$g(x) = 0 \quad x \in (0,1)$$

$$f(x) - f(x+1) = 0 \quad x \in (0,1)$$

$$\boxed{f(x) = f(x+1) \quad x \in (0,1)}$$

A function $f(x)$ is continuous in the interval $[0, 2]$. It is known that $f(0) = f(2) = -1$ and $f(1) = 1$. Which one of the following statements must be true?

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- ✓ A. There exists a y in the interval $(0, 1)$ such that $f(y) = f(y + 1)$
- B. For every y in the interval $(0, 1)$, $f(y) = f(2 - y)$
- C. The maximum value of the function in the interval $(0, 2)$ is 1
- ✓ D. There exists a y in the interval $(0, 1)$ such that $f(y) = -f(2 - y)$



$$f(0.5) = f(1.5)$$

$$g(x) = f(x) - f(x+1)$$

$$g(0) = f(0) - f(1) = f(0) - f(1)$$

$$g(1) = f(1) - \underbrace{f(2)}_{f(0)} = f(1) - f(0)$$

$$\underline{\underline{g(0)g(1) < 0}} \quad \& \quad g \text{ is cont}^n f^n \Rightarrow x \in (0,1)$$

$$f(y) = f(\underline{y+1}) \quad \underline{y \in (0,1)}$$

$$g(x) = 0$$

$$f(x) = f(x+1) \quad x \in (0,1)$$

$$f(y) = -f(2-y) \quad y \in (\underline{0}, \underline{1})$$

$$g(x) = f(x) + f(2-x)$$

$$g(0) = f(0) + f(2) = -2$$

$$g(1) = f(1) + f(1) = 2$$

$$\underline{g(0) \cdot g(1) < 0} \quad \& \quad g \text{ is a cont}^n f^n$$

$$g(x) = 0 \quad x \in (0, 1)$$

$$f(x) = -f(2-x) \quad x \in (0, 1)$$